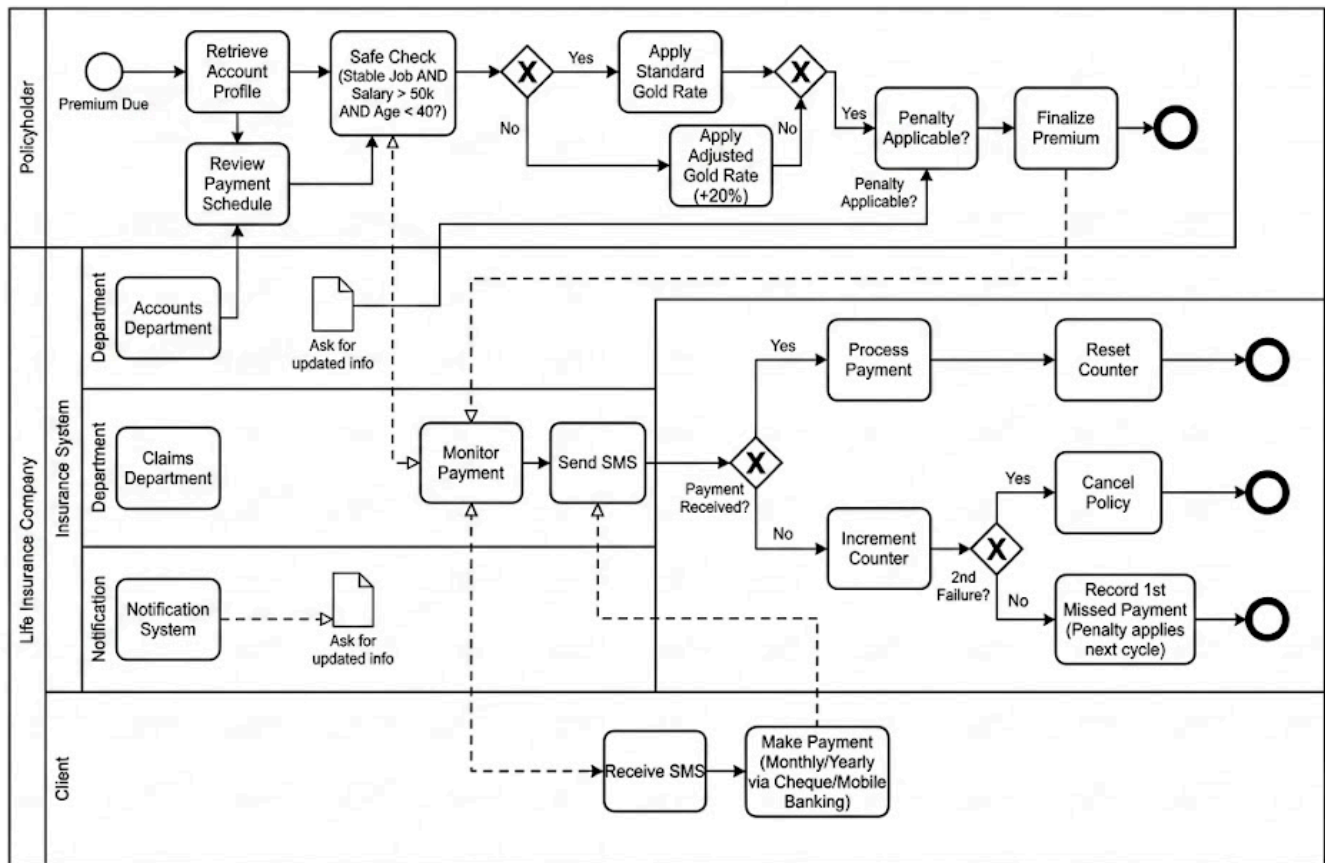
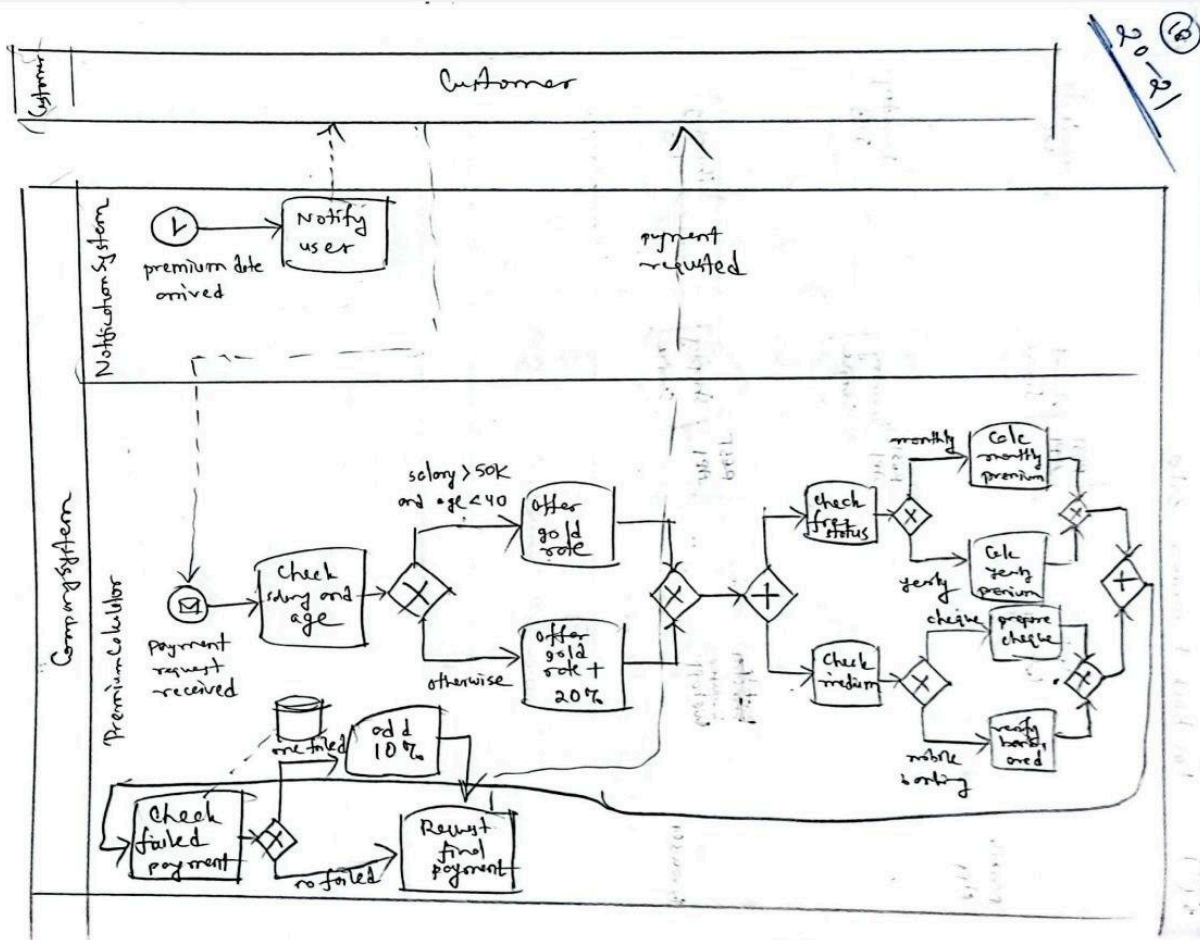


1. (a) You are designing software for a Life Insurance company. The requirements are as follows. If the client has a stable job with salary more than 50,000/= and his/her age is below 40 years, she/he is considered a safe customer and offered a Gold rate for premium payment. If any of these conditions are not made, the premium is 20% higher than the Gold rate. Clients may opt to pay premium installment monthly or yearly and via cheque or mobile banking. If a client fails to make a payment, a penalty of 10% of original premium is applied on the next premium. Before due date of premium, the client receives SMS mentioning the amount to pay. If she fails to pay two successive premiums, his/her policy is cancelled. Draw a BPMN diagram to design the premium calculation scenario discussed above. (14)





(b) What do you understand by Feature Envy, Refused Bequest and Cyclomatic Complexity type of Code Smell. Give an example of Inappropriate Intimacy in the context of Code Smell. (16)

Not in Syllabus

(c) What do you need to check about requirements collected for a software application? (5)

After collecting requirements for a software application, the following properties must be checked to ensure requirement quality and project success:

1. No Ambiguity

Requirements must be clear and precise so that developers and designers interpret them in the same way.

2. Accuracy and Completeness

All user goals, needs, and functionalities must be correctly captured without missing any important requirement.

3. Traceability

Each requirement should be traceable throughout development, design, implementation, and testing phases.

4. Verifiability

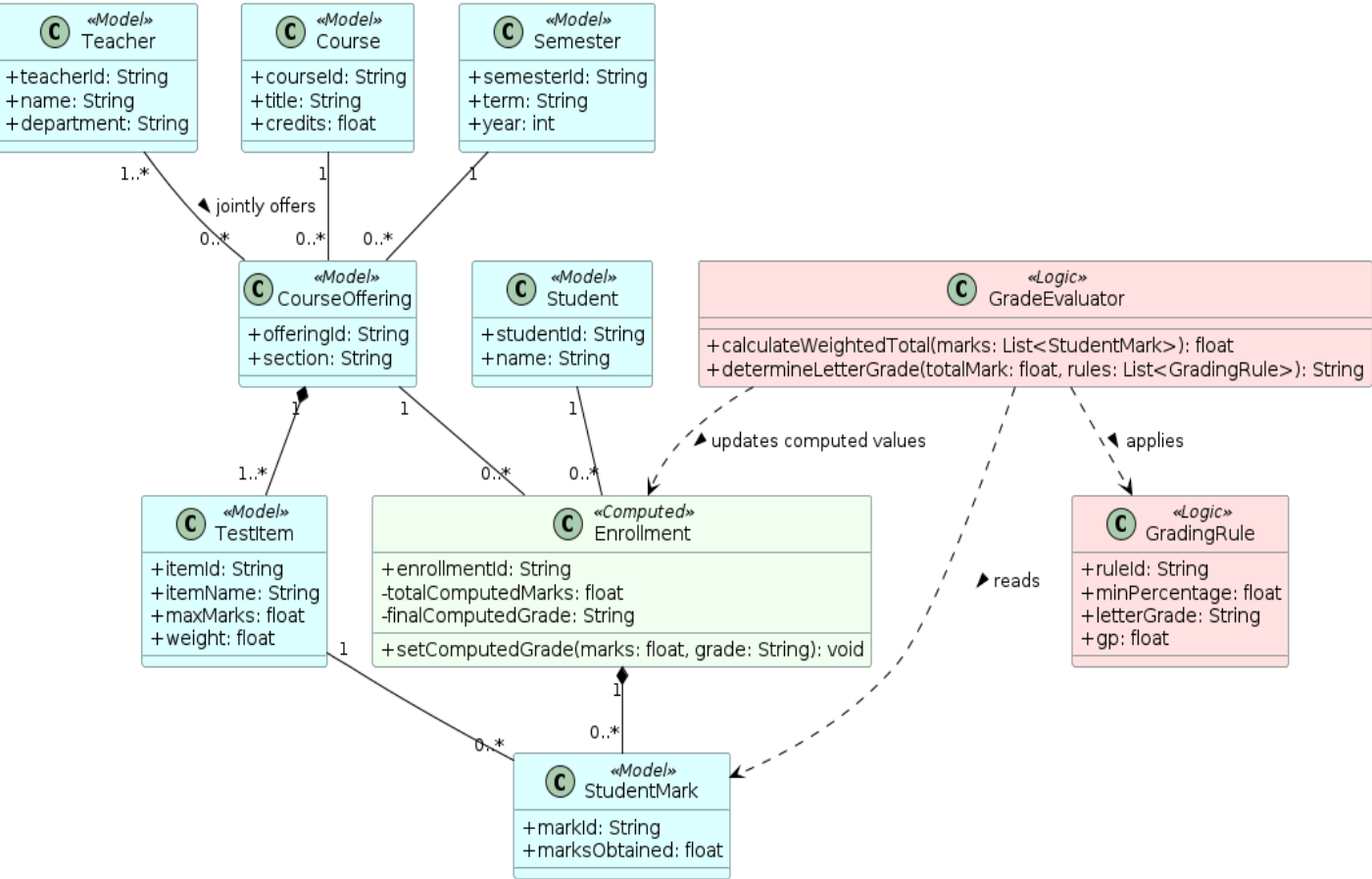
Requirements must be written in a testable manner so that the testing team can verify whether they are satisfied.

5. Consistency

Requirements should not conflict with one another and must remain logically consistent throughout the system.

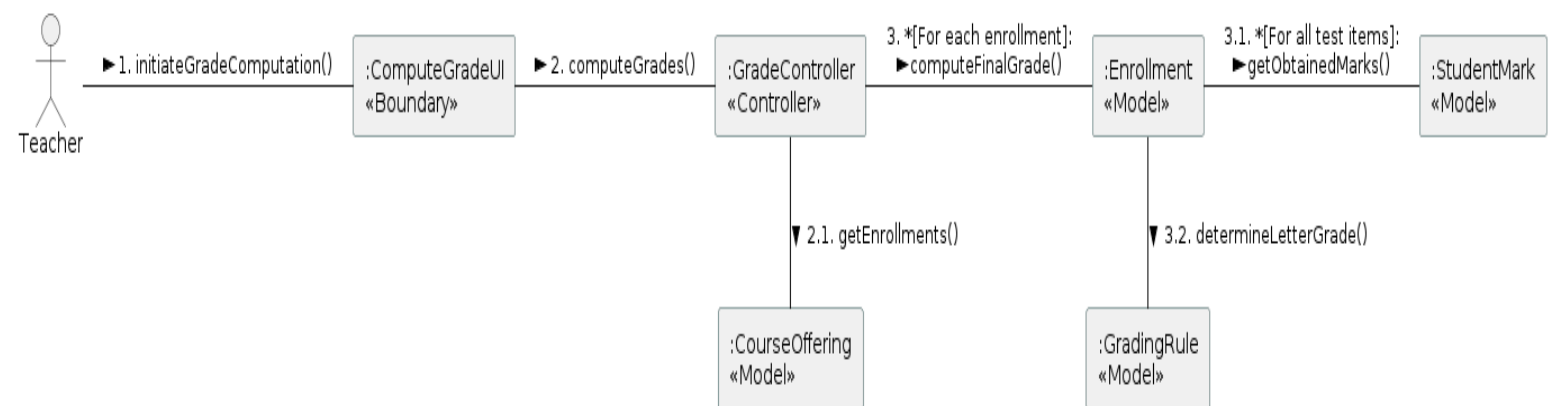
2. (a) In an educational institution, the teachers offer different course in different

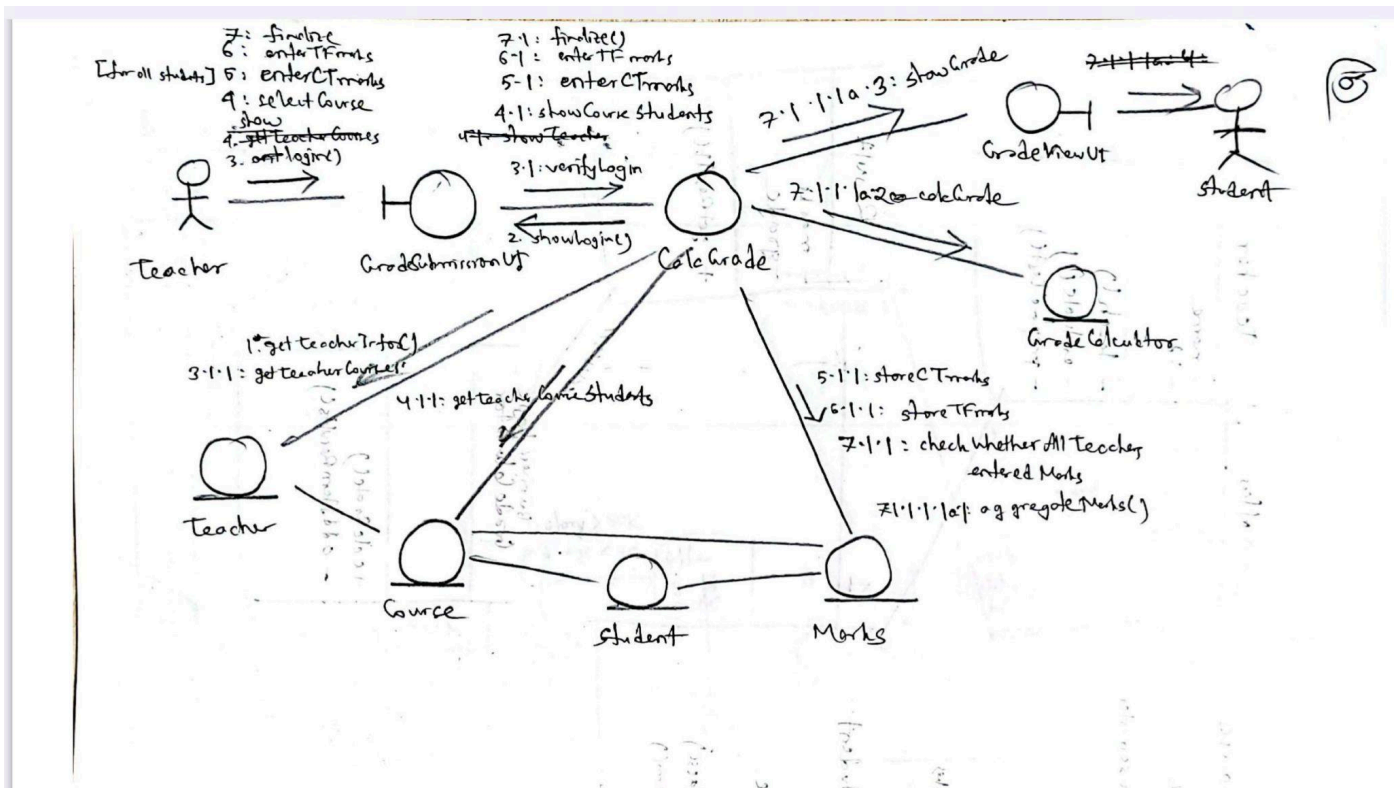
semesters. Some courses are jointly offered by multiple teachers. Students enroll in multiple offerings and receive grades at the end of the course. The grades are computed following some grading rule from the marks of different test items. Draw a Class Diagram to reflect the scenario discussed above. Apart from model classes, show the classes that would implement business logic and store computed values. You are not required to show any Boundary or Controller class. (13)



2. (a) In an educational institution, the teachers offer different courses in different semesters. Some courses are jointly offered by multiple teachers. Students enroll in multiple offerings and receive grades at the end of the course. The grades are computed following some grading rule from the marks of different test items. Draw a Class Diagram to reflect the scenario discussed above. Apart from model classes, show the classes that would implement business logic and store computed values. You are not required to show any Boundary or Controller class. (13)

(b) Draw Collaboration diagram for the scenario described in Q 2(a) to show the computation of grade. You are required to show relevant Boundary and Controller classes. (14)





(c) Give examples of 4 types of Non-Functional requirements for the example of the application to be developed for the scenario discussed in Q 2(a). (8)

For the educational institute's **grading and course enrollment system**, the following are examples of Non-Functional Requirements:

1. Performance Requirement

The system must calculate grades for at least **1000 students within 2 seconds** during course completion.

2. Security Requirement

Only authorized teachers can input or modify grades, and students' marks must remain encrypted in the database.

3. Availability Requirement

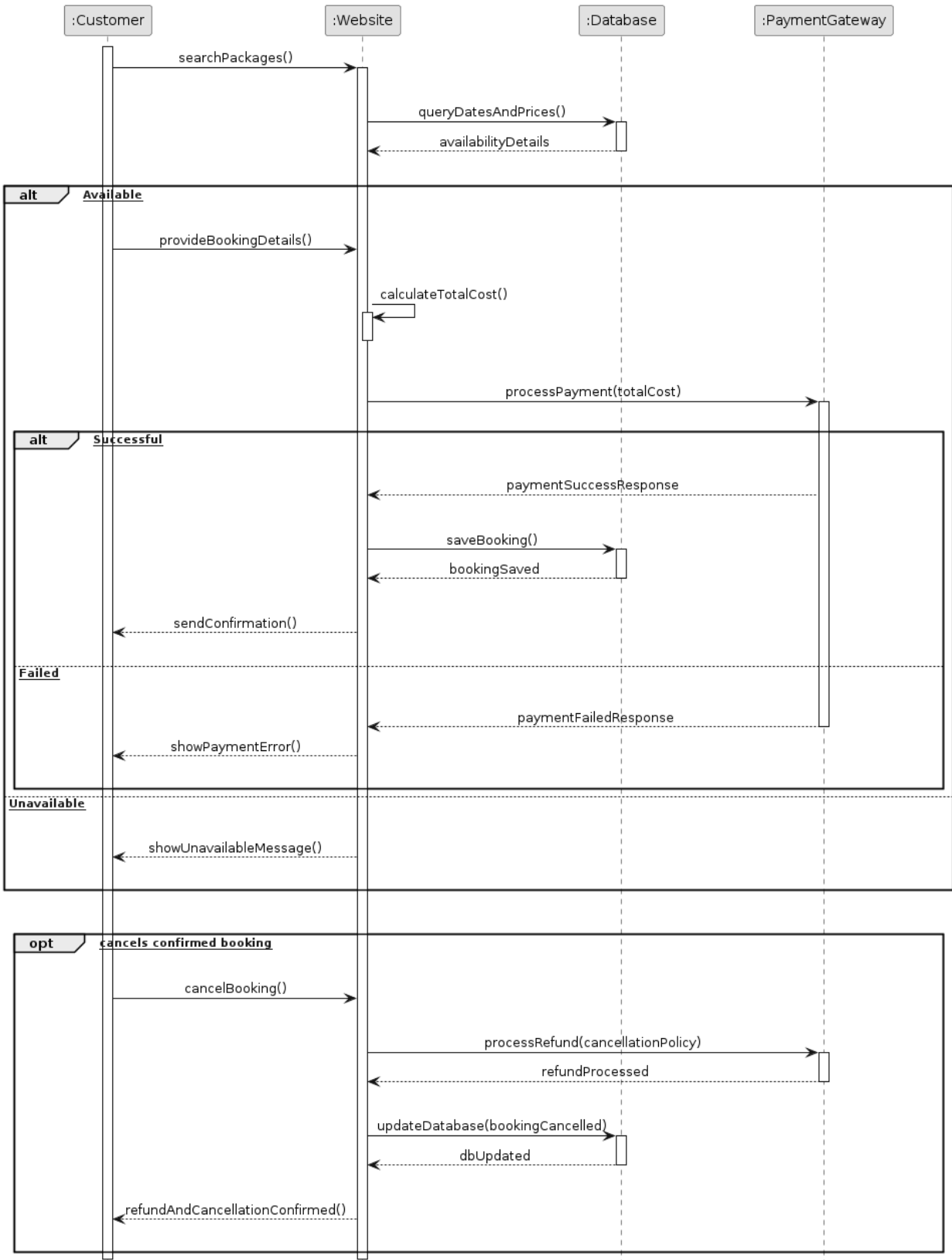
The system must maintain **99.99% uptime**, especially during course enrollment and result publication periods.

4. Usability Requirement

The grade submission interface should be simple and user-friendly so that teachers can use it without prior training.

3. (a) If you want to make a tour with a tourism operator, you have to select a package from their website. Each package presents a combination of site visit, hotels, food, activities. Depending on these components and number of days to spend, the price of the packages varies. A package can be customized for a group of tourists with size larger than 10 and price adjustment is made accordingly. The customer has to book a package with partial advance payment. The remaining payment is made prior to the tour to confirm all the components of the tour. If any activity is not available, the refund is made later. Draw a Sequence diagram to model the scenario discussed above.

(15)



(c) Show an example of "extends" in a Use Case diagram.

What is «extend»?

In a Use Case Diagram, the «extend» relationship is used when a use case performs an optional or conditional behavior that is not executed every time.

Extend = Optional additional functionality added to a base use case.

Important Rule

In an «extend» relationship, the arrow always points:

From Extension Use Case → To Base Use Case

Example Scenario

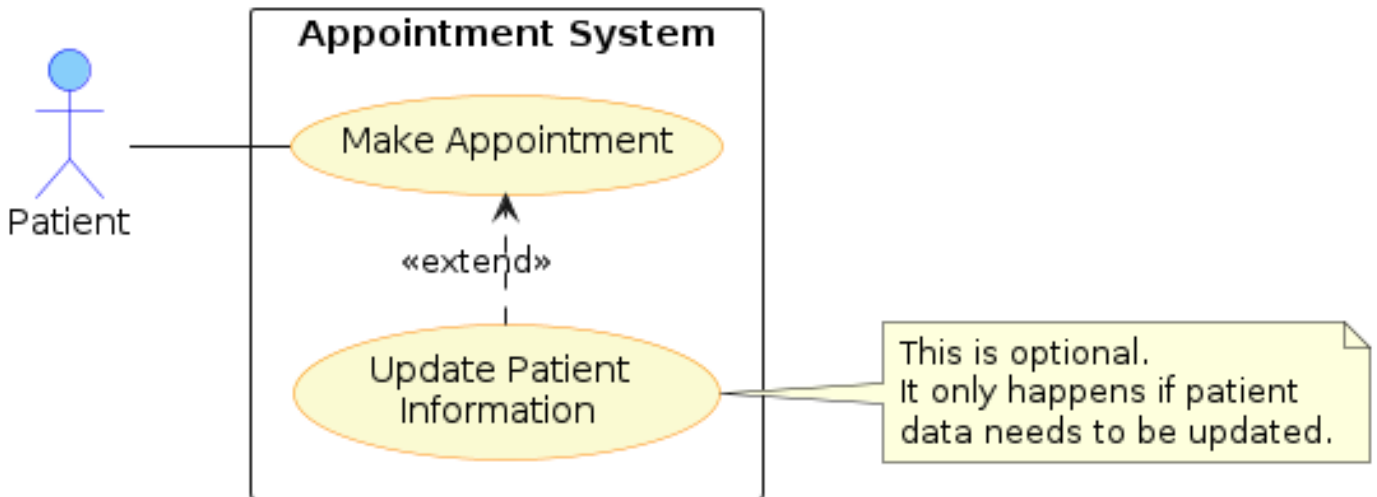
Consider a hospital management system where a patient wants to “Make Appointment.”

Normally, the patient can directly make the appointment.

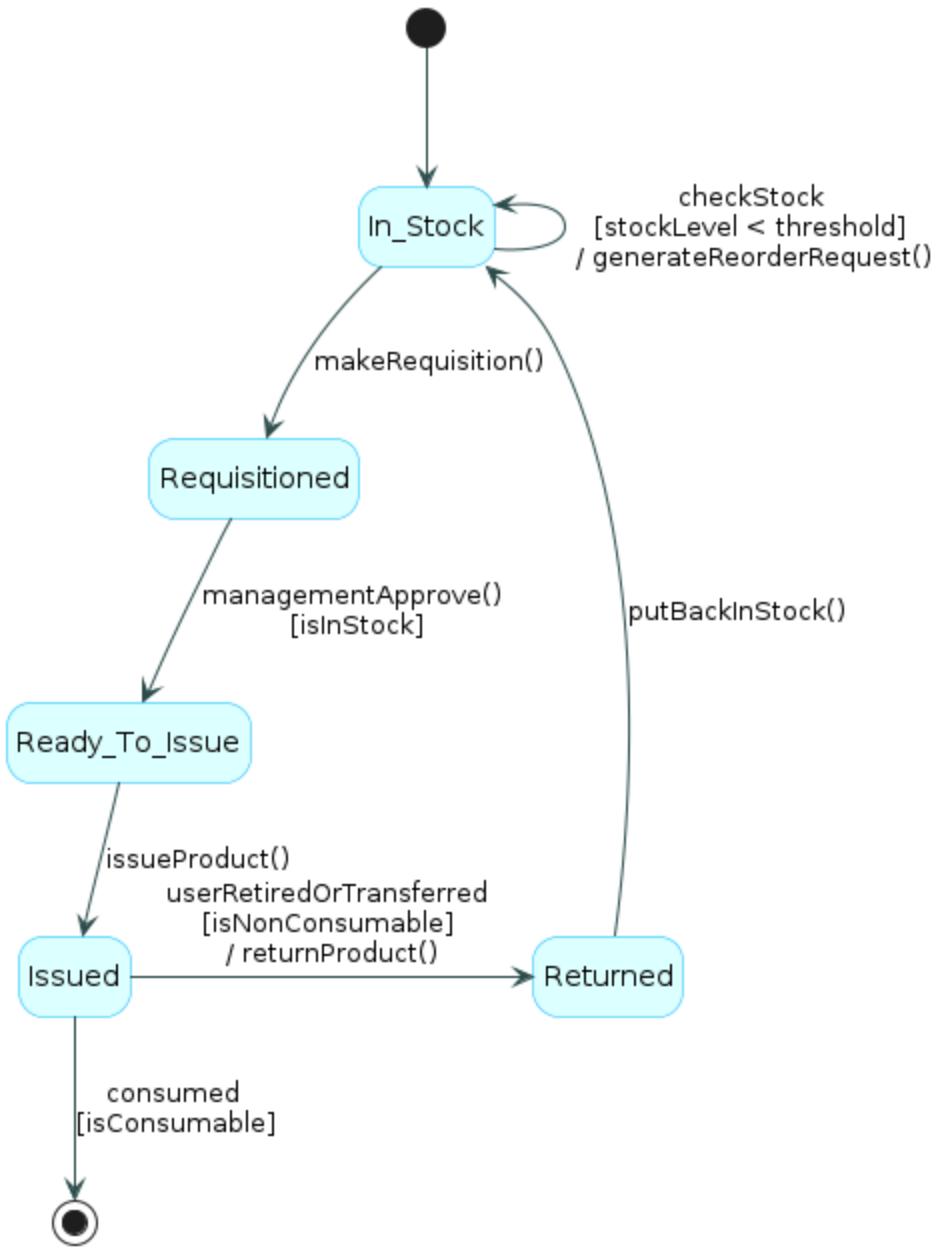
However, if the patient’s phone number or address has changed, the patient may need to “Update Patient Information.”

Since updating information is not mandatory every time, the use case:

“Update Patient Information” extends “Make Appointment.”



4(a) In an automatic Inventory Management System, an office stocks different products that are expected to be used by the users. When a user needs something, slhe makes a Requisition. If that is in the stock, the requisition is approved by the management and becomes ready to Issue. If the stock.level of a product reaches below a threshold, a reorder request is generated. If the product is not consumable, it has to be returned when the employee is retired or transferred. Design a State diagram to show different status of a product in the context discussed above.



4(c) What do you understand by relative importance of usability dimensions in the context of User Interface design? Discuss in the context of a Sales Management software.

4(c) Relative Importance of Usability Dimensions in UI Design

Usability in User Interface (UI) design mainly consists of three dimensions:

1. **Learnability** – how easy it is for users to learn the system.
2. **Efficiency** – how quickly users can perform tasks.
3. **Safety** – how well the system prevents errors and supports recovery.

The **relative importance** of these usability dimensions depends on:

- the type of **users** (novice or expert), and
- the nature of the **tasks** performed.

For novice users, **Learnability** is more important, while for expert users, **Efficiency** becomes the priority. In systems handling sensitive information, **Safety** is also critically important.

Context of Sales Management Software

1. Learnability

When a new sales representative joins the company, the software should be easy to understand and use.

Features such as:

- simple menus,
- intuitive dashboard,
- clear navigation,

help new users learn the system quickly with minimal training.

2. Efficiency

Sales Management Software is used regularly for:

- customer data entry,
- invoice generation,
- sales tracking,
- report preparation.

After continuous use, users become expert users. Therefore, the system should support:

- keyboard shortcuts,
- quick search,
- auto-fill forms,
- fast workflow navigation.

This increases productivity and reduces operational time.

3. Safety

Sales data is highly sensitive and valuable. Therefore, the interface must reduce the possibility of user errors.

Examples include:

- confirmation before deletion,
- Undo/Redo functionality,
- input validation,
- access control mechanisms.

These features help recover from mistakes and protect important customer data.

Conclusion

The importance of usability dimensions is not fixed; it varies depending on users and tasks. An effective Sales Management Software should balance **Learnability, Efficiency, and Safety** to ensure both ease of use and high productivity while maintaining data security and reliability.